22/07/2026

CODE-B
Phase-2

Aakash
Medical | IIT-JEE | Foundations

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FORTNIGHTLY TEST FOR NEET-2027 (XII PASSED)

MM: 720 **Test - 02** **Time: 180 Mins.**

Answers

1. (4)	37. (2)	73. (1)	109. (2)	145. (1)
2. (1)	38. (1)	74. (3)	110. (1)	146. (1)
3. (1)	39. (3)	75. (3)	111. (3)	147. (2)
4. (2)	40. (3)	76. (1)	112. (2)	148. (4)
5. (1)	41. (2)	77. (3)	113. (2)	149. (2)
6. (3)	42. (2)	78. (3)	114. (1)	150. (2)
7. (4)	43. (2)	79. (1)	115. (1)	151. (3)
8. (2)	44. (2)	80. (4)	116. (3)	152. (1)
9. (2)	45. (3)	81. (2)	117. (2)	153. (2)
10. (3)	46. (2)	82. (1)	118. (3)	154. (3)
11. (2)	47. (1)	83. (3)	119. (2)	155. (3)
12. (3)	48. (1)	84. (4)	120. (4)	156. (3)
13. (2)	49. (4)	85. (4)	121. (3)	157. (2)
14. (3)	50. (4)	86. (2)	122. (1)	158. (2)
15. (1)	51. (2)	87. (4)	123. (1)	159. (4)
16. (3)	52. (1)	88. (3)	124. (3)	160. (3)
17. (2)	53. (2)	89. (1)	125. (1)	161. (2)
18. (3)	54. (3)	90. (4)	126. (4)	162. (4)
19. (1)	55. (1)	91. (4)	127. (2)	163. (2)
20. (2)	56. (4)	92. (2)	128. (4)	164. (4)
21. (3)	57. (3)	93. (2)	129. (1)	165. (2)
22. (1)	58. (2)	94. (3)	130. (4)	166. (2)
23. (2)	59. (2)	95. (4)	131. (3)	167. (1)
24. (1)	60. (4)	96. (1)	132. (4)	168. (2)
25. (4)	61. (2)	97. (2)	133. (1)	169. (1)
26. (3)	62. (2)	98. (2)	134. (2)	170. (2)
27. (3)	63. (4)	99. (2)	135. (1)	171. (4)
28. (1)	64. (2)	100. (2)	136. (4)	172. (1)
29. (4)	65. (4)	101. (3)	137. (3)	173. (3)
30. (3)	66. (4)	102. (4)	138. (1)	174. (1)
31. (1)	67. (1)	103. (1)	139. (2)	175. (1)
32. (1)	68. (2)	104. (3)	140. (3)	176. (3)
33. (2)	69. (1)	105. (2)	141. (3)	177. (3)
34. (3)	70. (4)	106. (1)	142. (2)	178. (2)
35. (1)	71. (2)	107. (1)	143. (4)	179. (1)
36. (4)	72. (4)	108. (2)	144. (1)	180. (1)

Hints and Solutions

PHYSICS

(1) **Answer :** (4)
Solution:
$\vec{r} = 2t\hat{i} + 2t^2\hat{j}$
$\vec{a} = \frac{d^2\vec{r}}{dt^2} = 0 + 4\hat{j}$
$\vec{a} = \left(4\hat{j}\right)$ m/s^2

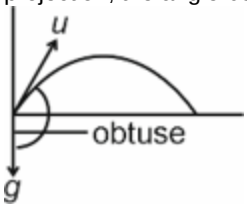
(2) **Answer :** (1)
Solution:
$a_1 = \frac{40-10}{5} = 6$ m/s^2
$a_2 = \frac{40-10}{1} = 30$ m/s^2
$\frac{a_1}{a_2} = \frac{6}{30} = \frac{1}{5}$

(3) **Answer :** (1)
Solution:
$y = x - 10x^2$
$y_{max} \rightarrow \frac{dy}{dx} = 0 \Rightarrow 0 = 1 - 20x$
$x = \frac{1}{20}m$
$y_{max} = \frac{1}{20} - 10\left(\frac{1}{20}\right)^2 = \frac{1}{40}$m

(4) **Answer :** (2)
Solution:
$R = \frac{2u_x u_y}{g} = \frac{2\times2\times3}{10}$
= 1.2 m

(5) **Answer :** (1)
Solution:
Projectile motion is a case of constant acceleration motion. On circular path angle between acceleration and velocity may be 90° or less than or greater than 90°.

(6) **Answer :** (3)
Solution:
If two particle projected with same speed at complimentary angle, then range of the particles will be same. At the point of projection, the angle between acceleration and velocity is obtuse.



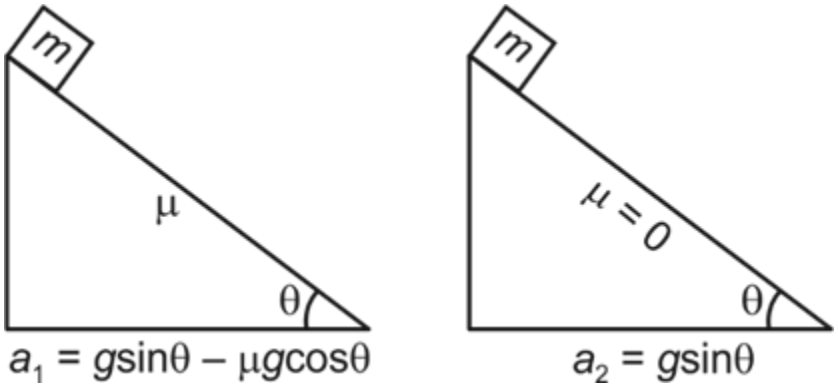
(7) **Answer :** (4)
Solution:
In uniform circular motion angular velocity and speed remains constant.

(8) **Answer :** (2)
Solution:
$\vec{A} - 2\vec{B} = (2\hat{i} + 3\hat{j} + 6\hat{k}) - 2(4\hat{i} - \hat{k})$
$= -6\hat{i} + 3\hat{j} + 8\hat{k}$

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$\frac{(\vec{A}-2\vec{B})}{\left|(\vec{A}-2\vec{B})\right|} = \frac{(-6\hat{i}+3\hat{j}+8\hat{k})}{\sqrt{36+9+64}} = \frac{(-6\hat{i}+3\hat{j}+8\hat{k})}{\sqrt{109}}$

(9) **Answer :** (2)
Solution:
Hint: Kinetic friction force on inclined plane (*f*) = μ*mg*cosθ
Sol:



$a_1 = g\sin\theta - \mu g\cos\theta$ $a_2 = g\sin\theta$

$\frac{t_1}{t_2} = 2 = \sqrt{\frac{g\sin\theta}{g\sin\theta-\mu g\cos\theta}}$

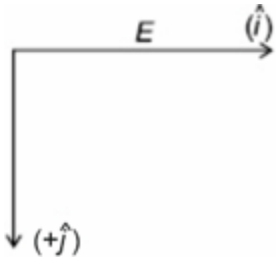
$\Rightarrow \quad 4 = \frac{\sin\theta}{\sin\theta-\mu\cos\theta}$

$\Rightarrow \quad 4\left(\frac{1}{\sqrt{2}} - \frac{\mu}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}}$

$\Rightarrow 4 - 4\mu = 1$

$\mu = \frac{3}{4}$

(10) **Answer :** (3)
Solution:



Let $\vec{v}_r = a\hat{i} + b\hat{j}$
In first case,
$\vec{v}_{rb} = \vec{v}_r - \vec{v}_b = \left(a-4\right)\hat{i} + b\hat{j}$
As rain appears to be falling vertically downward, (a – 4) = 0
⇒ a = 4
In second case,
$\vec{v}_b = 8\hat{i}$
$\vec{v}_{rb} = \left(a-8\right)\hat{i} + b\hat{j} = -4\hat{i} + b\hat{j}$
As rain appears to fall at 45° to the vertical.
tan 45° = $\frac{b}{4}$
or b = + 4
So, $\vec{v}_r = 4\hat{i} + 4\hat{j}$
$\left|\vec{v_r}\right| = 4\sqrt{2}$ m/s

(11) **Answer :** (2)

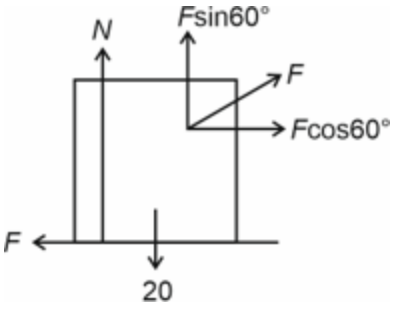
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Solution:

Hint & Sol.: $\left|\vec{a}\right| - \left|\vec{b}\right| \leq R \leq \left|\vec{a}\right| + \left|\vec{b}\right|$

Hence the resultant of two forces can't be more than the sum of magnitude of two forces.

(12) **Answer :** (3)
Solution:
μ = tanα
μ = tan30°
$= \left(\frac{1}{\sqrt{3}}\right)$



N = 20 – *F*sin60°
For (F)min
*F*cos60° = μ(20 – *F*sin60°)
$F = \frac{20\mu}{\cos 60^\circ + \mu \sin 60^\circ}$
$\Rightarrow F = \frac{20}{\sqrt{3}}\ \text{N}$

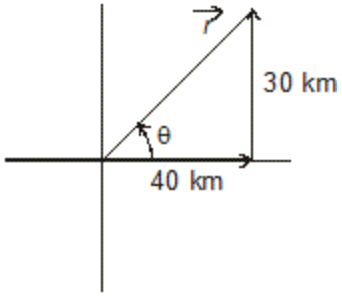
(13) **Answer :** (2)
Solution:
Since acceleration of box is due to static friction
$ma \leq \mu_s N = \mu_s mg$
$a \leq \mu_s g$
$a_{\max} = \mu_s g = 0.15 \times 10\ \text{ms}^{-2} = 1.5\ \text{ms}^{-2}$

(14) **Answer :** (3)
Solution:
Tension in string changes when string has own mass.

(15) **Answer :** (1)
Solution:
$T_1 = T_2$
$\frac{2\pi}{\omega_1} = \frac{2\pi}{\omega_2} \Rightarrow \omega_1 = \omega_2$

(16) **Answer :** (3)
Solution:
V_A sin 60º = V_B
$\Rightarrow \frac{V_A}{V_B} = \frac{2}{\sqrt{3}}$

(17) **Answer :** (2)
Solution:



$\vec{r} = 40\hat{i} + 30\hat{j}$

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$|\vec{r}|$= $\sqrt{(40)^2+(30)^2}$ = 50 km.

$\tan\theta$= $\frac{3}{4} \Rightarrow \theta = 37^\circ$

(18) Answer : (3)
Solution:

$\vec{v} = \vec{u} + \vec{a}\ t$

Step-1 $\left(4\hat{i}+6\hat{j}\right) = \left(2\hat{i}+2\hat{j}\right) + \vec{a} \times 2$

$\frac{\left(2\hat{i}+4\hat{j}\right)}{2} = \vec{a}$

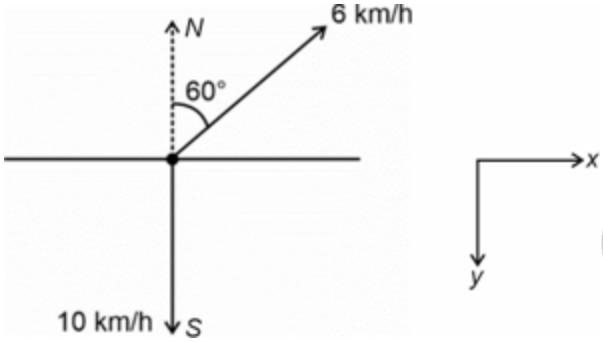
$\vec{a} = \left(\hat{i}+2\hat{j}\right)$ m/s^2

$\left|\vec{a}\right| = \sqrt{1+4} = \sqrt{5}$ m/s^2

(19) Answer : (1)
Solution:

Both $\vec{P}$ and $\vec{Q}$ are in same direction.

(20) Answer : (2)
Solution:



$\vec{V}_{M,G} = \vec{V}_{M,r} + \vec{V}_{r,G}$

$= 6\sin 60\hat{i} + \left[10 - 6\cos 60^\circ\right]\hat{j}$

$= 3\sqrt{3}\hat{i} + 7\hat{j}$

$\left|\vec{V}_{M,G}\right| = \sqrt{76}$ m/s and $\theta = \tan^{-1}\left[\frac{3\sqrt{3}}{7}\right]$ with south direction

(21) Answer : (3)
Solution:

$\tan\theta = \frac{v^2}{rg}$

$v = 180 \times \frac{5}{18} = 50\ \text{m/s}$

$\tan\theta = \frac{50\times 50}{500\times 10} = \frac{2500}{5000} = \frac{1}{2}$

$\theta = \tan^{-1}\left(\frac{1}{2}\right)$

(22) Answer : (1)
Solution:

p_1 = 20 Ns $\quad$ p_2 = 20 Ns

$|\Delta p| = |p_f - p_f| = 2p\sin\left(\frac{\theta}{2}\right)$

$= 2 \times 20 \times \sin\left(\frac{60^\circ}{2}\right) = 20\ \text{Ns}$

(23) Answer : (2)
Solution:

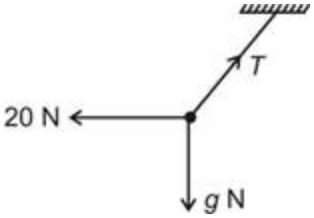
$T = mg - ma$

and $T = \frac{mg}{2} \Rightarrow \frac{Mg}{2} = Mg - Ma \Rightarrow Ma = \frac{Mg}{2} \Rightarrow a = \frac{g}{2}$

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(24) **Answer :** (1)
Solution:
$\mu mg\cos\theta > mg\sin\theta$
Friction = $mg\sin\theta = 50$ N

(25) **Answer :** (4)
Solution:

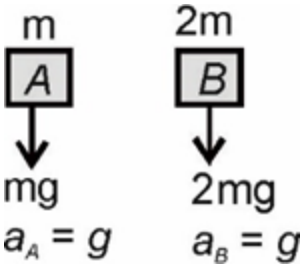


$T = \sqrt{(20)^2 + (10)^2}$
$T = \sqrt{500}$ N

(26) **Answer :** (3)
Solution:
Impulse equals to change in momentum.

(27) **Answer :** (3)
Solution:
$5\cos 45^\circ = \frac{5}{\sqrt{2}}$ km

(28) **Answer :** (1)
Solution:
In this case spring force is zero initially
F.B.D. of *A* and *B*



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(29) **Answer :** (4)
Solution:
For equilibrium N_A cos 60° + N_B cos 30° = Mg
and N_A sin 60° = N_B sin 30°
On solving $N_B = \sqrt{3}\, N_A$; $N_A = \frac{Mg}{2}$

(30) **Answer :** (3)
Solution:
Sol.: Static friction is a self-adjusting force, it can take values ranging from 0 to $\mu_S N$.
Limiting friction = $(F_S)_{max} = \mu_S N$
Kinetic friction = $\mu_k N$

(31) **Answer :** (1)
Solution:
In uniform circular motion, magnitude of acceleration remains constant.
A body with constant acceleration may have decreasing speed if velocity and acceleration are in opposite direction.

(32) **Answer :** (1)
Solution:
$2F = 5g \Rightarrow F = 25$ N

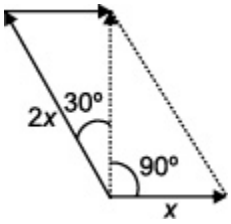
(33) **Answer :** (2)
Solution:
$\frac{v^2}{rg} \le \mu$

$v \le \sqrt{\mu rg}$

$v \le \sqrt{\frac{5}{10} \times 5 \times 10}$

$v \le 5$ m/s

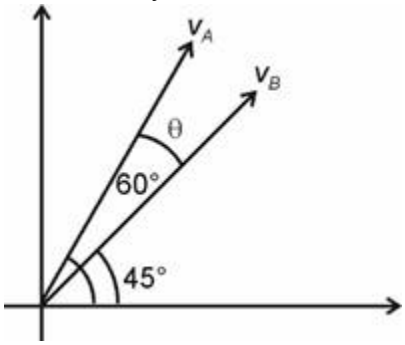
(34) Answer : (3)
Solution:
θ = 120º



(35) Answer : (1)
Solution:

$\bar{v}_A = 10\sqrt{3}\,\hat{i} + 30\hat{j}$

$\bar{v}_B = 20\,\hat{i} + 20\hat{j}$



∴ θ = 60° – 45°
= 15º.

(36) Answer : (4)
Solution:
f_A , max = 0.6 × 40 = 24 N
f_B, max = 0.6 × 60 = 36 N
$f_{total} \ge 5.5$ g
So no block will move in this situation.

(37) Answer : (2)
Solution:

$R = \frac{u^2 \sin^2\theta}{g} = \frac{98^2 \times \sin 90^\circ}{9.8} = \frac{(98)^2}{9.8} = 980$ m

Decrease in range = 980 – 950 = 30 m

(38) Answer : (1)
Solution:

$\sqrt{\frac{3^2}{25} + \frac{4^2}{25} + b^2} = 1$

$1 + b^2 = 1$
$b = 0$

(39) Answer : (3)
Solution:
At point of projection

$\vec{u} = u\cos\theta\hat{i} + u\sin\theta\hat{j}$

Final velocity $\vec{v} = u\cos\theta\hat{i} - u\sin\theta\hat{j}$

$\overrightarrow{\Delta v} = -2u\sin\theta\hat{j}$

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$\left|\overrightarrow{\Delta v}\right| = 2u\sin\theta$

(40) Answer : (3)
Solution:

As $\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}\left[10t^2 i + 5tj\right]$

$\vec{F}$= (20*t*i + 5*j*) N

at *t* = 2 s $\vec{F}$= (40*i* + 5*j*) N

(41) Answer : (2)
Solution:

$H_1 = \frac{u^2\sin^2\theta_1}{2g}$

$H_2 = \frac{u^2\sin^2\theta_2}{2g}$

$\frac{H_1}{H_2} = \frac{\sin^2\theta_1}{\sin^2\theta_2} = \frac{3}{1}$

$\frac{\sin\theta_1}{\sin\theta_2} = \sqrt{3} = \tan 60^\circ = \frac{\sin 60^\circ}{\cos 60^\circ} = \frac{\sin 60^\circ}{\sin 30^\circ}$

$\theta_1 = 60^\circ$, $\theta_2 = 30^\circ$

When speed of *A* is doubled

$\frac{R_1}{R_2} = \frac{\frac{4u^2\sin 2\theta_1}{g}}{\frac{u^2\sin 2\theta_2}{g}}$

$= \frac{4\sin 120^\circ}{\sin 60^\circ} = \frac{4\times\frac{\sqrt{3}}{2}}{\frac{\sqrt{3}}{2}} = 4$

(42) Answer : (2)
Solution:

- Frictional force always oppose relative motion between the surface of contact
- Coefficient of static friction is greater than coefficient of kinetic friction for given pair of surfaces
- Friction force is non-conservative force in nature.

(43) Answer : (2)
Solution:

$F = \frac{mv^2}{r} \Rightarrow \frac{v^2}{r} = \text{constant} \quad \Rightarrow v \propto \sqrt{r}$

$\frac{v_1}{v_2} = \sqrt{\frac{r_1}{r_2}} = \frac{\sqrt{3}}{2}$

(44) Answer : (2)
Solution:

With maximum height H_1 and H_2

For same horizontal range

$R = 4\sqrt{H_1 \times H_2} = 4 \times \sqrt{16 \times 9}$

= 4 × 12 = 48 m

(45) Answer : (3)
Solution:

a_t = 4 m/s^2

$a_c = \frac{v^2}{r} = \frac{(u+a_t t)^2}{r}$

$a_c = \frac{a_t^2 t^2}{r}$

$a_c = \frac{16\times 20\times 20}{100} = 64\ \text{ms}^{-2}$

$\therefore a_{net} = \sqrt{a_c^2 + a_t^2}$

$\frac{a_t}{a_{net}} = 1 : \sqrt{257}$

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CHEMISTRY

(46) **Answer :** (2)
Solution:
• 6th excited state = $n_2 = 6 + 1 = 7$
• 1st excited state = $n_1 = 1 + 1 = 2$
• Electron is making a transition from 7th to 2nd, energy level so zero lines in UV region.

(47) **Answer :** (1)
Solution:

$\nu = \frac{c}{\lambda}$

$\tilde{v} = \frac{\nu}{c} = \frac{7.5 \times 10^{14}}{3 \times 10^{8}}$

$= 2.5 \times 10^{6}\ m^{-1}$

(48) **Answer :** (1)
Solution:
Diffraction and interference are explained by wave nature of electromagnetic radiations.

(49) **Answer :** (4)
Solution:
- Number of electrons ejected is proportional to the intensity of light.
- The kinetic energy of the ejected electrons increases with increase of frequency of light.

(50) **Answer :** (4)
Solution:
The correct order of wavelength of given electromagnetic waves is radiowaves > Microwave > IR > UV

(51) **Answer :** (2)
Solution:
Power of bulb = 30 watt = $30\ Js^{-1}$
Energy of photon = $\frac{hc}{\lambda} = \frac{6.6\times10^{-34}\times3\times10^{8}}{1320\times10^{-9}} = 1.5 \times 10^{-19}$ J
Number of photons emitted per second = $\frac{30}{1.5\times10^{-19}} = 20 \times 10^{19} = 2 \times 10^{20}$

(52) **Answer :** (1)
Solution:
$E = \frac{hc}{\lambda}$
$E \propto \frac{1}{\lambda}$
$\frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1}$
$\frac{\lambda_1}{\lambda_2} = \frac{E_1}{E_2} = \frac{10}{5} = 2$
$\lambda_1 = 2\lambda_2$.

(53) **Answer :** (2)
Hint:
$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$
Solution:
For He^+ ion
$\frac{1}{\lambda} = R \times (2)^2\left(\frac{1}{1} - \frac{1}{2^2}\right)$
or, $\frac{1}{\lambda} = R \times 4 \times \frac{3}{4}$
or, $\lambda = \frac{1}{3R} = x$
or, $R = \frac{1}{3x}$
For Li^{2+}

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$\frac{1}{\lambda} = R \times (3)^2 \left(\frac{1}{2^2} - \frac{1}{3^2}\right)$

$= \frac{1}{3x} \times 9 \times \frac{5}{36} = \frac{5}{12x}$

$\lambda = \frac{12x}{5}$ Å

(54) **Answer :** (3)
Hint:
Photoelectric effect is the emission of electrons from metal surface when light of suitable energy falls on metal surface.
Solution:
Stark effect and Zeeman effect are the splitting of spectral lines under the influence of electric field and magnetic field respectively.

(55) **Answer :** (1)
Solution:
Energy of one photon = $\frac{hc}{\lambda} \quad (\lambda = 300\ \text{nm})$

For one mole photons, $E = \frac{hc}{\lambda} \times N_A$

$E = \frac{6.626 \times 10^{-34} \times 3 \times 10^8 \times 6.023 \times 10^{23}}{300 \times 10^{-9}}$

$E = 3.99 \times 10^5$ J mol^{-1}

Kinetic energy = 1.68×10^5 J mol^{-1}

W_0 = E – K.E.

$= 3.99 \times 10^5 - 1.68 \times 10^5$

$= 2.31 \times 10^5$ J mol^{-1}

(56) **Answer :** (4)
Solution:
• Atomic number (Z) = Number of protons = number of electrons in neutral atom
Z = 57 = Number of protons
57 = Number of electrons
Number of neutrons = 138 – 57 = 81

(57) **Answer :** (3)
Hint:
n = 4 and *l* = 2 is 4*d*
Solution:
d orbitals are five in number so one *d* orbital will occupy maximum 2 electrons.

(58) **Answer :** (2)
Solution:

n	***l***	**Orbital**
2	1	2*p*
3	2	3*d*
1	0	1*s*
3	1	3*p*

(59) **Answer :** (2)
Solution:

$\lambda = \frac{h}{mv}$

$= \frac{6.6 \times 10^{-34}}{20 \times 10^{-6} \times 500}$

$\lambda = 6.6 \times 10^{-32}\text{m}$

(60) **Answer :** (4)
Solution:

Orbital angular momentum $= \sqrt{l(l+1)}\,\frac{h}{2\pi}$

$= \sqrt{2}\,\frac{h}{2\pi}$

(61) **Answer :** (2)
Hint:
The value of m varies from –ℓ to +ℓ
Solution:

For $\ell = 2$, m = –2, –1, 0, +1, +2
So for $\ell = 2$, m = +3 is not possible

(62) **Answer :** (2)
Hint:
According to Pauli's exclusion principle; no two electrons in an atom can have the same set of four quantum numbers.
Solution:
It follows only Pauli's exclusion principle. It violates Hund's rule of maximum multiplicity and Aufbau principle.

(63) **Answer :** (4)
Hint:
Chromium and copper have exceptional electronic configuration.
Solution:
$_{29}Cu = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$
$_{24}Cr = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$
$_{25}Mn = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^2$
$_{26}Fe = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6\ 4s^2$

Element	Unpaired electron(s)
Cu	1
Mn	5
Cr	6
Fe	4

(64) **Answer :** (2)
Solution:
$\Delta V = 1\%$of 100 = $\frac{1\times100}{100} = 1$m/s
$\Delta x = \frac{h}{4\pi m\Delta V}$
$= \frac{6.6\times10^{-34}\times1000}{4\times3.14\times10\times1} = 5.25\times10^{-33}$m

(65) **Answer :** (4)
Hint:
For one electron species, energy of subshell depends only on the value of n.
Solution:
For 3rd shell, No. of degenerates orbitals = $n^2 = 3^2 = 9$.

(66) **Answer :** (4)
Solution:
For the 2 electrons in same orbital the value of n, l and m_ℓ are same so these can be distinguished by only spin quantum number.

(67) **Answer :** (1)
Hint:
Atomic number 36 is of Krypton (Kr) and 48 is of cadmium (Cd) and both are 5th period elements.
Solution:
Z [49] $5s^2\ 5p^1$, 5th period
group number = 10 + 2 + 1 = 13
Z [92] $5f^3\ 6d^1\ 7s^2$, 7th period
All *f*-block elements are in 3rd group
Z [38] $5s^2$, 5th period and 2nd group
Z [57] $5d^1\ 6s^2$, 6th period and 3rd group

(68) **Answer :** (2)
Hint:

Electronic configuration	Element
$1s^2 2s^2$	Be
$1s^2 2s^2 2p^1$	B
$1s^2 2s^2 2p^3$	N
	O

$1s^2 2s^2 2p^4$	

Solution:
Due to stable electronic configuration of 'Be' & 'N', they have higher first ionization enthalpy than B and O respectively

B < Be < C < O < N < F

(801 899 1086 1314 1402 1681) $\Delta_i H$ / kJ mol^{-1}

(69) Answer : (1)
Solution:

Element	Δ_{eg} H(kJ mol^{-1})
Br	– 325
F	– 328
S	– 200
Se	– 195

(70) Answer : (4)
Solution:
The oxidation state of Al is +3 and covalency is 6.
x – 1 + 0 × 5 = +2
x = +3

(71) Answer : (2)
Solution:

	Element	Electronegativity value on pauling scale
1.	Be	1.5
2.	B	2.0
3.	C	2.5
4.	S	2.5
5.	Al	1.5
6.	P	2.1
7.	Na	0.9
8.	Mg	1.2

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(72) Answer : (4)
Solution:
Atomic number (19) = K
Atomic number (11) = Na
K has lower first ionisation enthalpy than Na.

(73) Answer : (1)
Solution:
Na_2O → basic
Cl_2O_7 → acidic
Al_2O_3 → Amphoteric
CO → Neutral

(74) Answer : (3)
Solution:
Among isolelectronic species the ionic radii follows the order.
$Mg^{2+} < Na^{+} < F^{-} < O^{2-}$.

(75) Answer : (3)
Hint:
Second electron gain enthalpy of an element is always positive.
Solution:
$O(g) \rightarrow O^{-}(g)$ (First $\Delta_{eg}H$)

$O^{-}(g) \rightarrow O^{2-}(g)$ (Second $\Delta_{eg}H$) (Always endothermic)

(76) **Answer :** (1)
Solution:

Element	Atomic radii/pm
Li	152
Be	111
Na	186
Mg	160

(77) **Answer :** (3)
Solution:
Metallic character increases down the group and decreases along a period.

Element	Electronegativity value
P	2.1
Si	1.8
Be	1.5
Mg	1.2
Na	0.9

P < Si < Be < Mg < Na

(78) **Answer :** (3)
Solution:
Arsenic and Tellurium are metalloids.

(79) **Answer :** (1)
Solution:
$Na^{+} + 1e^{-} \rightarrow Na$ $\Delta H = -5.1$ eV

(80) **Answer :** (4)
Solution:

$\lambda = \frac{h}{\sqrt{2m\,K.E.}}$ where h and m (mass) are constants

$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{K_2}{K_1}}$

$K_2 = 3K_1$

$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{3K_1}{K_2}}$

$\frac{\lambda_1}{\lambda_2} = \sqrt{3}$

$\lambda_2 = \frac{\lambda_1}{\sqrt{3}}$

(81) **Answer :** (2)
Solution:
5p → n + l = 6
5s → n + l = 5
4p → n + l = 5
3d → n + l = 5
(n + l) value increases the energy of orbital also increases. For same value of (n + l), higher is the value of n, higher is the energy of orbital.
So, correct order is 5p > 5s > 4p > 3d

(82) **Answer :** (1)
Solution:
Number of radial nodes = n – l – 1
For 4p, n = 4, l = 1
∴ No. of radial nodes = 4 – 1 – 1 = 2

(83) **Answer :** (3)

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Solution:
There is large difference between 3rd and 4th ionisation energy.
$\therefore$ +3 oxidation state will be more stable and the formula of halide is MX_3.

(84) Answer : (4)
Solution:
Group 17 elements are called halogens.

(85) Answer : (4)
Solution:

Kinetic Energy = – Total Energy = $\frac{13.6 \times Z^2}{n^2}$

= $\frac{13.6 \times 4}{9}$

= 6.04 eV

(86) Answer : (2)
Solution:

$r = 0.53 \times \frac{n^2}{Z}$

$r_{Be^{3+}} = 0.53 \times \frac{2^2}{4}$ $\left[1^{st}\text{excited state n = 1 + 1 = 2}\right]$

r = 0.53 Å

(87) Answer : (4)
Hint:
de Broglie wavelength, $\lambda = \frac{h}{mv}$

$mvr = \frac{nh}{2\pi}$
Solution:
In 4th orbit, angular momentum $= \frac{4h}{2\pi}$

$= \frac{2h}{\pi}$

$n\lambda = 2\pi r$

$\lambda = \frac{2\pi \times a_0 \times n}{Z}$

$\left(\text{as } r = a_0 \frac{n^2}{z}\right)$

$\lambda = 6\pi a_0$

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(88) Answer : (3)
Solution:
We know that

$\Delta E = 13.6 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$ where $n_2 > n_1$, $n_2 = 4$ to $n_1 = 3$ will give least energetic photon.

(89) Answer : (1)
Solution:
Gallium is also known as Eka-aluminium.

(90) Answer : (4)
Solution:
After removal of first electron, removal of second electron requires more energy and hence second ionisation enthalpy is more than that of first ionisation enthalpy.

BOTANY

(91) Answer : (4)
Hint:
Identification, nomenclature and classification are the basic to both taxonomy and systematics.
Solution:

Systematics takes into account evolutionary relationships (phylogenetic relationships) along with identification, nomenclature and classification.

(92) Answer : (2)
Hint:
ICNB (International code of Nomenclature of Bacteria) assigns rules for scientific naming of bacteria.
Solution:
ICBN (International Code for Botanical Nomenclature) assigns rules for scientific naming of plants.

(93) Answer : (2)
Hint:
Lion and leopard are two different species but both belong to the genus, *Panthera*.
Solution:
Plant family Convolvulaceae is included in the order Polymoniales, mainly on the basis of floral characters.

(94) Answer : (3)
Hint:
In S phase, DNA replication takes place.
Solution:
G_1 phase – Interval between mitosis and initiation of DNA replication
G_2 phase – Cytoplasmic growth takes place.
M phase – In prophase, condensation of chromatin fibre begins.

(95) Answer : (4)
Hint:
Mitosis helps to repair the lining of gut.
Solution:
Meiocytes (PMC) are the cells that undergo meiosis to produce gametes.

(96) Answer : (1)
Hint:
Mammalia is a class.
Solution:
Felidae and canidae are placed in the order, carnivora.

(97) Answer : (2)
Solution:
Linnaeus used *Systema Naturae* as the title of his publication.

(98) Answer : (2)
Hint:
The taxonomic hierarchy is as follows:

Kingdom
↑
Phylum or Division
↑
Class
↑
Order
↑
Family
↑
Genus
↑
Species

Solution:
The correct depiction of sequential order of these taxa in ascending order is as follows:
Poaceae → Poales → Monocotyledonae → Angiospermae → Plantae

(99) Answer : (2)
Hint:
Different kinds of organisms are referred to as biodiversity of the earth.
Solution:
Biodiversity is the number and various kinds of organisms found on earth. It stands for the variability found among living organisms inhabiting this world.

(100) Answer : (2)
Solution:
Karyokinesis is the division of nucleus.

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(101) Answer : (3)

Hint:

Nomenclature is the naming of living organisms.

Solution:

Classification is the process by which anything is grouped into convenient categories based on some easily observable characters.

(102) Answer : (4)

Solution:

Wheat and mango both belongs to kingdom plantae and division Angiospermae.

Wheat is a monocot and mango is a dicot.

(103) Answer : (1)

Hint:

R.H. Whittaker has proposed five kingdom classification. Carolus Linnaeus has proposed binomial nomenclature.

Solution:

Ernst Mayr has proposed biological species concept.

Ernst Haeckel has proposed three kingdom classification.

(104) Answer : (3)

Solution:

In the quiescent stage (G_0), cells remain metabolically active but no longer proliferate.

(105) Answer : (2)

Solution:

In oocytes of some vertebrates, diplotene lasts for months or years.

(106) Answer : (1)

Hint:

Diakinesis is the final stage of prophase-I of meiosis-I.

Solution:

Diakinesis is marked by the terminalisation of chaismata.

(107) Answer : (1)

Hint:

Meiosis I is responsible for reduction in chromosomes number.

Solution:

In metaphase I,

- Bivalents align at equator.
- Two equatorial plates are formed.
- Each chromosome has two chromatids.
- Each chromosome is attached to the spindle fibre from one pole only.

(108) Answer : (2)

Solution:

Centrifugal cytokinesis is seen in plants and its assisted by cell plate.

(109) Answer : (2)

Solution:

If number of bivalents = 20, then total number of chromosomes = 40 in diploid cell and 80 chromatids.

(110) Answer : (1)

Hint:

Phylum is used for animals.

Solution:

Dicotyledonae – Class
Arthropoda – Phylum
Diptera – Order
Angiospermae – Division

(111) Answer : (3)

Solution:

Interphase lasts about more than 95% of the duration of cell cycle.

(112) Answer : (2)

Solution:

Cell growth results in disturbing the ratio between the nucleus and the cytoplasm. It therefore becomes essential for the cell to divide to restore the nucleo-cytoplasmic ratio.

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(113) Answer : (2)

Hint:

Chromatids do not separate in meiosis I.

Solution:

In meiosis I, only homologous chromosomes separate and each chromosome has two chromatids.
Thus, in interkinesis, there is no need of DNA duplication as duplicated DNA forms sister chromatids.

(114) Answer : (1)

Hint:

Recombinase enzyme is involved in crossing over.

Solution:

Crossing over occurs in pachytene. Crossing over is exchange of genetic material between two homologous chromosomes which is an enzyme mediated process and the enzyme involved is recombinase.

(115) Answer : (1)

Hint:

Liquid endosperm in coconut is multinucleated in nature

Solution:

In some organisms karyokinesis is not followed by cytokinesis as a result of which multinucleated condition arises.

(116) Answer : (3)

Hint:

The two asters together with the spindle fibres form mitotic apparatus.

Solution:

Each centrosome radiates out microtubules called asters.
Centromere is the primary constriction of the chromosome and is not a part of the mitotic apparatus.

(117) Answer : (2)

Hint:

A cell cycle involves interphase and M phase. During the division of a cell, DNA replication and cell growth take place.

Solution:

The sequence of events by which a cell duplicates its genome, synthesizes the other constituents of the cell and eventually divides into two daughter cells is termed as 'cell cycle'.

(118) Answer : (3)

Solution:

At the time of cytoplasmic division (cytokinesis) organelles like mitochondria and plastids get distributed between the two daughter cells.

(119) Answer : (2)

Solution:

Centrosomes radiate some microtubules called asters.

(120) Answer : (4)

Solution:

The duplication of semi-autonomous organelles, such as mitochondria and chloroplasts, occurs during the G_2 phase , where the cell prepares for mitosis.

(121) Answer : (3)

Solution:

A typical eukaryotic cell cycle is illustrated by human cells in culture. These cells divide once in approximately every 24 hours.
Yeast progresses through the cell-cycle in only about 90 minutes.

(122) Answer : (1)

Solution:

Prophase I is divided into five sub phases.
The longest phase of reduction division is prophase I.

(123) Answer : (1)

Solution:

Chiasmata or X-shaped structures become visible during diplotene stage.

(124) Answer : (3)

Solution:

Higher plants lack centrioles but still they make spindle fibres.

(125) Answer : (1)

Hint:

Best stage to study shape of chromosomes is anaphase.

Solution:

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Metaphase is the best stage to study the morphology of chromosomes.

(126) Answer : (4)
Hint:
In animal cells, centriole duplication takes place in S phase.
Solution:
In animal cells, during the S phase, DNA replication begins in the nucleus, and the centriole duplicates in the cytoplasm.

(127) Answer : (2)
Hint:
Meiosis-I is the actual reductional division
Solution:
During anaphase-I, homologous chromosome pair separate.

(128) Answer : (4)
Solution:
Figure (a) and (b) represent mitotic anaphase and metaphase stage of cell division respectively. Fig. (c) and (d) represent meiotic anaphase-I and metaphase-I stage respectively.

(129) Answer : (1)
Solution:
Congression of bivalents to the equator plate during Metaphase-I of Meiosis.

(130) Answer : (4)
Solution:
The correct sequence of the stages of cell cycle is:
$G_1 \rightarrow S \rightarrow G_2 \rightarrow M$

(131) Answer : (3)
Solution:
Hint: In S phase, there is duplication of DNA not chromosomes.
Sol.: In G_2 phase the chromosome number will remain constant that is 40.

(132) Answer : (4)
Solution:
Hint : Leptotene, zygotene, diplotene and diakinesis are the substages of prophase-I of meiosis.
Sol. : Both in prophase of mitosis and prophase I of meiosis, condensation of chromatin materials occurs. Pairing of homologous chromosomes and crossing over occur in prophase-I of meiosis.

(133) Answer : (1)
Hint:
The class of *Solanum, Petunia* and *Datura* is same *i.e.* Dicotyledonae.
Solution:
Three different genera *Solanum*, *Petunia* and *Datura* are placed in the family Solanaceae.

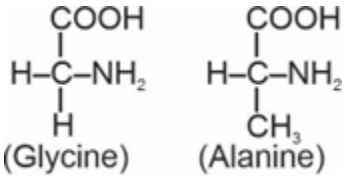
(134) Answer : (2)
Hint:
Binomial epithet includes generic name, specific epithet and author citation.
Solution:
The second component in the biological name denotes the specific epithet.

(135) Answer : (1)
Solution:
Human beings have always been interested in knowing more and more about the various kinds of organisms, particularly with reference to their own use. In early days, human beings needed to find sources for their basic needs of food, clothing and shelter. Hence, the earliest classifications were based on the ‘uses’ of various organisms.

ZOOLOGY

(136) Answer : (4)
Solution:
Lecithin is a nitrogen containing phospholipid.

(137) Answer : (3)
Solution:



(138) Answer : (1)

Solution:

Conjugated enzyme is known as holoenzyme which is made up of apoenzyme and coenzyme.

(139) Answer : (2)

Solution:

When two or more polypeptide subunits come together, it is called quaternary structure. Alpha helix or beta-pleated sheet represent secondary structures of proteins.

(140) Answer : (3)

Hint:

Prosthetic group

Solution:

There are three kinds of co-factors–prosthetic groups, co-enzymes and metal ions.

Prosthetic groups are organic compounds and are distinguished from other co-factors in that they are tightly bound to the apoenzyme. For example, in peroxidase and catalase, which catalyse the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and it is a part of active site of the enzyme. Zinc is a co-factor for the proteolytic enzyme carboxypeptidase.

(141) Answer : (3)

Hint:

Select the basic amino acid.

Solution:

Tyrosine, tryptophan and phenylalanine are aromatic amino acids.

Lysine is a basic amino acid with an additional ($–NH_2$) group.

(142) Answer : (2)

Solution:

A general rule of thumb is that the rate of reaction doubles or decreases by half for every 10°C change in either direction. Therefore, the rate of reaction becomes one fourth of the initial if the temperature had been decrease by 20°C.

(143) Answer : (4)

Hint:

Cellulose is a polymer of glucose.

Solution:

Haemoglobin is made up of 2 parts *i.e.*, haem and globin. Globin is a protein part and contains peptide bonds. Nucleosides are formed by nitrogenous bases and pentose sugar.

(144) Answer : (1)

Solution:

Based on Chargaff's rule

A + G = T + C

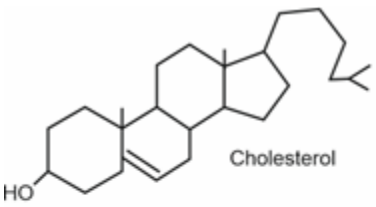
[A] = [T] & [G] = [C]

C = 20%, A = 30% and T = 30%

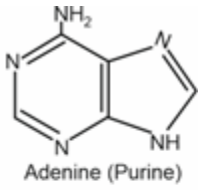
(145) Answer : (1)

Solution:

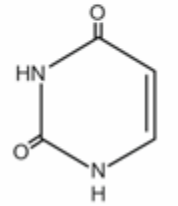
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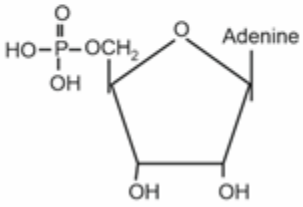
Cholesterol



Adenine (Purine)



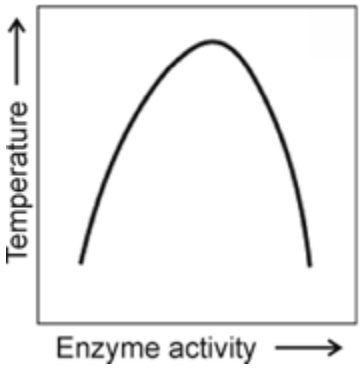
Uracil (Pyrimidine)



Adenylic acid

(146) Answer : (1)

Solution:



(147) Answer : (2)

Hint:

Catalyses breakdown of hydrogen peroxide to water and oxygen

Solution:

Prosthetic groups are organic compounds and are distinguished from other co-factors in that they are tightly bound to the apoenzyme. For example, in peroxidase and catalase, which catalyse the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and it is a part of the active site of the enzyme.

• Zinc is a co-factor for the enzymes, carboxypeptidase and carbonic anhydrase.

(148) Answer : (4)

Solution:

Usually four types of interactions cooperate in stabilizing the tertiary structure. These are disulphide bonds, hydrophobic interactions, hydrogen bonds and ionic interactions.

(149) Answer : (2)

Hint:

An enzyme present in plants.

Solution:

Chitin is present in the exoskeleton of arthropods. Collagen is the most abundant protein in the animal world while Ribulose Bisphosphate Carboxylase-Oxygenase (RuBisCO) is the most abundant protein in whole of the biosphere.

(150) Answer : (2)

Hint:

Connects adjacent nucleotides

Solution:

In nucleic acids, one ester bond is present on each side of phosphate group therefore, it is called phosphodiester bond. Peptide bond links amino acids in a polypeptide. Glycosidic bond links individual monosaccharides in a polysaccharide. Hydrogen bonds stabilize the tertiary structure of proteins.

(151) Answer : (3)

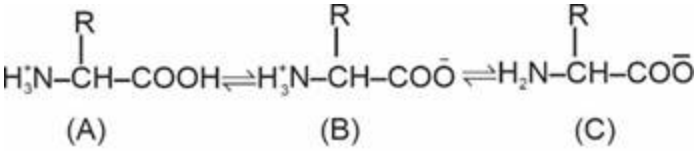
Solution:

Hydrolases are the enzymes which catalyse breakage of bond present between the glycerol and fatty acids in a molecule of triglyceride and their enzyme commission number starts with 3.

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(152) Answer : (1)
Solution:
Ribozyme is a ribonucleic acid (RNA) enzyme that catalyzes a chemical reaction.

(153) Answer : (2)
Solution:



B is called zwitterionic form

(154) Answer : (3)
Solution:
A competitive inhibitor binds reversibly with the enzyme. Its effect can be overcome by increasing the concentration of the substrate.
Competitive inhibition refers to a situation where an inhibitor competes with the substrate for binding at the substrate binding site of an enzyme. It increases the K_m value but does not affect the maximum velocity (V_{max}) of the reaction.

(155) Answer : (3)
Solution:
The lungs are situated in the thoracic chamber which is anatomically an air-tight chamber. The thoracic chamber is formed dorsally by the vertebral column, ventrally by the sternum, laterally by the ribs and on the lower side by the dome-shaped diaphragm.

(156) Answer : (3)
Hint:
Function performed by alveoli
Solution:
The part starting with the external nostrils up to the terminal bronchioles constitute the conducting part whereas the alveoli and their ducts form the respiratory or exchange part of the respiratory system.
The conducting part transports the atmospheric air to the alveoli, clears it from foreign particles, humidifies and also brings the air to body temperature. Exchange part is the site of actual diffusion of O_2 and CO_2 between blood and atmospheric air.

(157) Answer : (2)
Solution:
All mammals respire by lungs.

(158) Answer : (2)
Hint:
Largest class in the animal kingdom
Solution:
Lower invertebrates like sponges, coelenterates, flatworms, *etc.*, exchange O_2 with CO_2 by simple diffusion over their entire body surface.

(159) Answer : (4)
Hint:
Forced exhalation is an active process.
Solution:
Forceful exhalation is due to contraction of internal inter-costal and abdominal muscles.

(160) Answer : (3)
Hint:
Opening of the larynx is called glottis.
Solution:
During deglutition, food can enter from pharynx into the larynx. This is prevented by closure of glottis with epiglottis.

(161) Answer : (2)
Hint:
Number of external nostrils in humans
Solution:
Human lungs are covered by a double layered pleura with pleural fluid between them. It reduces friction on the lung surface.

(162) Answer : (4)
Solution:

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During normal quiet inspiration, intra-pulmonary pressure is less than the atmospheric pressure.

(163) Answer : (2)
Solution:
Diffusion of gases is directly proportional to surface area.

(164) Answer : (4)
Solution:
The trachea, primary, secondary and tertiary bronchi and initial bronchioles are supported by incomplete cartilaginous rings.

(165) Answer : (2)
Solution:
The contraction of external inter-costal muscles lifts up the ribs and the sternum causing an increase in the volume of thoracic chamber in the dorso-ventral axis.

(166) Answer : (2)
Hint:
It is also called sound box.
Solution:
Vocal cords present inside the larynx are responsible for sound production. A part of pharynx is common passage for both food and air. Trachea is also known as wind pipe.

(167) Answer : (1)
Solution:
The nasal chamber opens into the pharynx, a portion of which is the common passage for food and air. The pharynx opens through the larynx region into the trachea.

(168) Answer : (2)
Solution:
Breathing is exchange of gases between atmospheric air and lungs.

(169) Answer : (1)
Hint:
Intrapulmonary pressure decreases during inspiration.
Solution:
In normal respiration, contraction of diaphragm and external intercostal muscles increases the volume of thoracic cavity leading to a decrease in intrapulmonary pressure.

(170) Answer : (2)
Solution:
A spirometer is a device used to measure the volume of air inhaled and exhaled by the lungs. However, it cannot measure certain lung volumes directly. Residual Volume (RV) is the volume of air remaining in the lungs after a maximal exhalation, and it cannot be measured by a spirometer. So, any respiratory capacity that contains RV will not be measured by spirometer.
FRC = ERV + RV and TLC = TV + ERV + IRV + RV.

(171) Answer : (4)
Solution:
Tidal Volume (TV): Volume of air inspired or expired during a normal respiration. It is approx. 500 mL.
Inspiratory Reserve Volume (IRV): Additional volume of air, a person can inspire by a forcible inspiration. This averages 2500 mL to 3000 mL.
Expiratory Reserve Volume (ERV): Additional volume of air, a person can expire by a forcible expiration. This averages 1000 mL to 1100 mL.
Residual Volume (RV): Volume of air remaining in the lungs even after a forcible expiration. This averages 1100 mL to 1200 mL.

(172) Answer : (1)
Hint:
It cannot be measured by a simple spirometer.
Solution:
Volume of air remaining in the lungs even after a forcible expiration is called the residual volume. This averages around 1100 mL to 1200 mL.
Tidal volume – Volume of air inspired or expired during a normal respiration. It is approximately 500 mL.
IRV – A person can inspire by a forcible inspiration (2500-3000 mL).
ERV – A person can expire by a forcible expiration (1000-1100 mL).

(173) Answer : (3)
Hint:
Divide minute volume by tidal volume
Solution:
Minute volume = Tidal volume × Breathing rate

9000 mL/minute = 500 mL × Breathing rate

∴ Breathing rate = $\frac{9000\ \text{mL}\ \text{perminute}}{500\ \text{mL}}$

= 18 times per minute

(174) Answer : (1)

Solution:

Here X represents VC

TLC = VC + RV

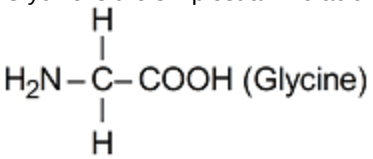
= IC + ERV + RV

= TV + IRV + ERV + RV

(175) Answer : (1)

Solution:

Glycine is the simplest amino acid which is optically inactive.



(176) Answer : (3)

Solution:

3 fatty acids + 1 glycerol $\rightarrow$

Triglyceride (fat)

(177) Answer : (3)

Hint:

Remain liquid at room temperature.

Solution:

Vegetable oils have low melting point, so they remain liquid at room temperature; lipids are not strictly considered as biomacromolecule due to their low molecular weight. Vegetable oils are examples of triglycerides.

(178) Answer : (2)

Solution:

Glottis is an opening of larynx.

(179) Answer : (1)

Hint:

Hemoglobin is tetrameric compound.

Solution:

Adult human haemoglobin represents quaternary structure of a protein, contains two subunits of α type polypeptide chains and two subunits of β type polypeptide chains. So, it is a tetrameric protein molecule with iron in Fe^{+2} state.

(180) Answer : (1)

Solution:

Cytosine + Ribose sugar ⇒ Cytidine

A nucleoside consists of a nitrogenous base attached to a sugar molecule.

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